i calculate the entropy change in the system for a reaction, ΔS_{system}, given entropy data

j use the expression $\Delta S_{surroundings} = -\Delta H/T$ to calculate the entropy change in the surroundings and hence ΔS_{total}

k demonstrate an understanding that the feasibility of a reaction depends on the balance between ΔS_{system} and $\Delta S_{surroundings}$, and that at higher temperatures the magnitude of $\Delta S_{surroundings}$ decreases and its contribution to ΔS_{total} is less. Reactions can occur as long as ΔS_{total} is positive even if one of the other entropy changes is negative

l demonstrate an understanding of and distinguish between the concepts of thermodynamic stability and kinetic inertness

m calculate ΔS_{system} and $\Delta S_{surroundings}$ for the reactions in 4.4g to show that endothermic reactions can occur spontaneously at room temperature

n define the term *enthalpy of hydration* of an ion and use it and lattice energy to calculate the enthalpy of solution of an ionic compound

o demonstrate an understanding of the factors that affect the values of enthalpy of hydration and the lattice energy of an ionic compound

p use entropy and enthalpy of solution values to predict the solubility of ionic compounds.

4.5 Equilibria

Knowledge of the concepts introduced in Unit 2, *Topic 2.9: Chemical equilibria* will be assumed and extended in this topic.

Students will be assessed on their ability to:

a demonstrate an understanding of the term *dynamic equilibrium* as applied to states of matter, solutions and chemical reactions

b recall that many important industrial reactions are reversible

c use practical data to establish the idea that a relationship exists between the equilibrium concentrations of reactants and products which produces the equilibrium constant for a particular reaction, e.g. data on the hydrogen-iodine equilibrium

d calculate a value for the equilibrium constant for a reaction based on data from experiment, e.g. the reaction of ethanol and ethanoic acid (this can be used as an example of the use of ICT to present and analyse data), the equilibrium $Fe^{2+}(aq) + Ag^{+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$ or the distribution of ammonia or iodine between two immiscible solvents

e construct expressions for K_c and K_p for homogeneous and heterogeneous systems, in terms of equilibrium concentrations or equilibrium partial pressures, perform simple calculations on K_c and K_p and work out the units of the equilibrium constants

f demonstrate an understanding that when ΔS_{total} increases the magnitude of the equilibrium constant increases since $\Delta S_{total} = R\ln K$

g apply knowledge of the value of equilibrium constants to predict the extent to which a reaction takes place

h relate the effect of a change in temperature on the value of ΔS_{total}.